An explicit weighted localization bound for a signed prime–gamma form

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27 October 2026
Preprint version 1.0

Abstract

We study a real even quadratic form consisting of a nonlocal gamma energy, the complete prime-power correlation sum, a local mass term, and a rank-one pole term. In an explicit Jacobi-theta weighted space, we construct a cubic angle partition of unity and bound the complete signed localization defect without losing any exterior gamma energy. For an interior radius $R \geq 1$ and $S = e^{2R}$, the defect is at most $9000(1+\log S)S^{-5/2}e^{-\pi S/2}$ times the squared weighted norm. We also cancel the deterministic finite-prime main phase against the full pole exactly and prove a complete weighted remainder estimate. Matching the cutoff support to $\log N$ gives a finite-N localization formula with no second interior truncation error. The theta inputs and all constants are proved directly. The remaining interior form is infinite-dimensional and signed; no positivity theorem or conclusion about the Riemann hypothesis is obtained.

Keywords. Weighted quadratic forms; nonlocal localization; prime powers; Jacobi theta function; signed correlation.
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1 The form and the localization statement

The issue addressed here is quantitative localization, not the sign of the localized form. An exact partition-of-unity expansion leaves a signed cross-edge defect. The rapid decay of a theta weight makes that defect small, and a cubic angle taper supplies an additional fourth-order gain. The proof keeps the complete prime-power sum and the pole term throughout. Generic partition identities and exterior localization are the underlying mechanisms; no historical novelty or priority claim is made for them.

For $t > 0$ define

$$\vartheta(t) = \sum_{j\in\mathbb{Z}} e^{-\pi j^2 t}, \qquad G(x) = e^{x/2}\vartheta(e^{2x}), \qquad \kappa(x) = \tfrac{1}{2}\big(G''(x) - G(x)/4\big).$$

Put $C(x) = \cosh(x/2)$, $W(x) = \kappa(x)/(2C(x))$, and

$$X = L^2_{\mathrm{even}}(\mathbb{R}, W^{-1}\,\mathrm{d}x; \mathbb{R}), \qquad N_X(u) = \|u\|_X^2 = \int_{\mathbb{R}} \frac{|u(x)|^2}{W(x)}\,\mathrm{d}x. \tag{1.1}$$

Appendix A proves, independently of any zero-location statement, that W is smooth, positive and even, and that

$$W(x) \le 40\exp(4|x| - \pi e^{2|x|}), \qquad \int_{\mathbb{R}} 2C(x)\kappa(x)\,\mathrm{d}x = 1. \tag{1.2}$$

Initially let u belong to the real even core $\mathcal{D}_0 = C_c^\infty(\mathbb{R};\mathbb{R}) \cap X$. We use

$$F_u(\xi) = \int_{\mathbb{R}} u(x)e^{-i\xi x}\,\mathrm{d}x, \qquad H_u(s) = \int_{\mathbb{R}} u(x)u(x+s)\,\mathrm{d}x, \qquad M(u) = \|u\|_2^2. \tag{1.3}$$

The correlation is real and even. With Λ denoting the von Mangoldt function, so that $\Lambda(p^k) = \log p$ for every prime p and integer $k \ge 1$ and $\Lambda(n) = 0$ otherwise, set

$$\rho(s) = \frac{e^{-s/2}}{1-e^{-2s}}, \qquad t_\Gamma(u) = \int_0^\infty \rho(s)\|u(\cdot+s) - u\|_2^2\,\mathrm{d}s, \tag{1.4}$$

$$a_n = \frac{\Lambda(n)}{\sqrt{n}}, \qquad p_N(u) = 2\sum_{2\le n\le N} a_n H_u(\log n), \tag{1.5}$$

$$c_\Gamma = \log\pi + \gamma + \frac{\pi}{2} + 3\log 2, \qquad \mathrm{Pole}(u) = 2\left|\int_{\mathbb{R}} C(x)u(x)\,\mathrm{d}x\right|^2. \tag{1.6}$$

Here γ is Euler's constant. The elementary bounds $0 < \gamma < 1$, $\log\pi < 2$, $\pi/2 < 2$ and $\log 2 < 1$ give $0 < c_\Gamma < 8$. The first bounds follow by comparing harmonic sums with the integral of $1/t$; the others follow from $3 < \pi < 4$ and $2 < e < 3$. The pole equals $2|F_u(i/2)|^2$ because u is real and even. Define

$$q_N(u) = t_\Gamma(u) - c_\Gamma M(u) - p_N(u) + \mathrm{Pole}(u), \qquad q_\infty(u) = t_\Gamma(u) - c_\Gamma M(u) - p_\infty(u) + \mathrm{Pole}(u). \tag{1.7}$$

The complete series p_∞ is an X-bounded form, as proved below. The gamma energy is once-oriented: its shifts run over $s > 0$ only. The factor two in (1.5) retains both prime orientations.

Our cutoffs are completely specified. Let

$$b(t) = \begin{cases} 0, & t \le 0, \\ 3t^2 - 2t^3, & 0 < t < 1, \\ 1, & t \ge 1, \end{cases} \quad \theta_R(x) = \frac{\pi}{2}b(|x| - R), \quad \chi_R(x) = \cos\theta_R(x), \quad \eta_R(x) = \sin\theta_R(x). \tag{1.8}$$

They are even C^1 functions, not necessarily smooth at their transition endpoints. They satisfy $\chi_R^2 + \eta_R^2 = 1$, $\chi_R = 1$ on $[-R, R]$ and $\chi_R = 0$ outside $[-R-1, R+1]$. Section 2 admits their products with core tests by compact H^1 approximation.

Theorem 1.1 (Explicit signed localization)**.** *For every $R \ge 1$ and $u \in \mathcal{D}_0$,*

$$|q_\infty(u) - q_\infty(\chi_R u) - t_\Gamma(\eta_R u)| \le \varepsilon_R N_X(u), \tag{1.9}$$

where, with $S = e^{2R}$,

$$\varepsilon_R \le 9000(1 + \log S)S^{-5/2}e^{-\pi S/2}. \tag{1.10}$$

An explicit, sharper expression for ε_R is given in (5.8)*. In particular the coefficient of the retained exterior energy $t_\Gamma(\eta_R u)$ is exactly one.*

The conclusion is a vanishing error in the original weighted norm. The single uncontrolled term $q_\infty(\chi_R u)$ has finite physical support but is still infinite-dimensional. A finite number of prime shifts on that support does not reduce its gamma energy to a finite-dimensional problem. Theorem 1.1 does not assert that this term is positive.

2 Weighted correlation bounds and the form domain

Write $\beta = \sqrt{W}$. The envelope in (1.2) implies that X embeds continuously in ordinary L^2 and that every $u \in X$ has every finite exponential L^1 moment: for $a \geq 0$,

$$\int_{\mathbb{R}} e^{a|x|}|u(x)|\,\mathrm{d}x \leq N_X(u)^{1/2}\left(\int_{\mathbb{R}} e^{2a|x|}W(x)\,\mathrm{d}x\right)^{1/2} < \infty.$$

Consequently all pole and correlation integrals used below are absolutely justified on X.

Lemma 2.1 (Complete weighted tails)**.** *For $s \geq 0$ and $u \in X$,*

$$\beta(x)\beta(x+s) \leq h(s) := 40e^{2s-\pi e^s}, \qquad |H_u(s)| \leq h(s)N_X(u). \tag{2.1}$$

For every integer $N \geq 1$,

$$|p_\infty(u) - p_N(u)| \leq r_N N_X(u), \qquad r_N = 80\sum_{n>N}(\log n)n^{3/2}e^{-\pi n}. \tag{2.2}$$

Moreover $|p_\infty(u)| \leq r_1 N_X(u)$ and

$$0 \leq \mathrm{Pole}(u) \leq \tfrac{1}{2}N_X(u). \tag{2.3}$$

For $A \geq 2$ define

$$\tau_A = 80\int_A^\infty t^{3/2}e^{-\pi t}\,\mathrm{d}t \leq \frac{80A^{3/2}e^{-\pi A}}{\pi - 3/(2A)}. \tag{2.4}$$

Proof. Put $a = |x|$, $b = |x+s|$ and $v = a + b \geq s$. The arithmetic–geometric mean inequality and the decrease of $2v - \pi e^v$ for $v \geq 0$ give

$$\beta(x)\beta(x+s) \leq 40e^{2v-(\pi/2)(e^{2a}+e^{2b})} \leq 40e^{2v-\pi e^v} \leq 40e^{2s-\pi e^s}.$$

Writing $u = \beta f$ and applying ordinary translation-invariant Cauchy–Schwarz proves the correlation estimate. At $s = \log n$, use $\Lambda(n) \leq \log n$ and sum every $n > N$ to obtain (2.2). No prime power is omitted. Finally

$$\int_{\mathbb{R}} C^2 W\,\mathrm{d}x = \frac{1}{2}\int_{\mathbb{R}} C\kappa\,\mathrm{d}x = \frac{1}{4},$$

which proves (2.3). Concavity of $\log t$ gives $t^{3/2} \leq A^{3/2}e^{3(t-A)/(2A)}$ for $t \geq A$; integration proves (2.4). □

The Fourier multiplier of the gamma form is

$$d_\Gamma(\xi) = 2\int_0^\infty \rho(s)(1-\cos(\xi s))\,\mathrm{d}s, \qquad t_\Gamma(u) = \frac{1}{2\pi}\int_{\mathbb{R}} d_\Gamma(\xi)|F_u(\xi)|^2\,\mathrm{d}\xi. \tag{2.5}$$

Tonelli and Plancherel justify the nonnegative integral. Expanding $\rho(s) = \sum_{j\geq 0} e^{-(2j+1/2)s}$ gives

$$0 \leq d_\Gamma(\xi) \leq \xi^2\int_0^\infty s^2\rho(s)\,\mathrm{d}s = 2\xi^2\sum_{j\geq 0}(2j+1/2)^{-3} \leq 18\xi^2.$$

Indeed the $j = 0$ term is 16, while the others sum to at most $\frac{1}{4}\sum_{j\geq 1} j^{-3} \leq 3/8$.

If v is real, even, compactly supported and in $H^1(\mathbb{R})$, convolution with even compact smooth approximate identities gives $v_j \in \mathcal{D}_0$ converging in H^1, with supports in one common compact interval. Since W has a positive minimum on that interval, $v_j \to v$ in X, and

$$t_\Gamma(v_j - v) \le 18\|(v_j - v)'\|_2^2 \longrightarrow 0. \tag{2.6}$$

The prime, pole and local mass terms are continuous in X. Thus all form values converge. This admits $\chi_R u$ and $\eta_R u$ to the completion of the original core in the norm $(N_X + t_\Gamma)^{1/2}$. No identification with an unspecified maximal form domain is required. In particular the gamma form itself has not been declared bounded on X.

3 Exact main-phase cancellation

For an integer $N \ge 2$, put $L = \log N$ and

$$P_N(\xi) = 2 \sum_{2 \le n \le N} a_n \cos(\xi \log n), \tag{3.1}$$

$$M_N(\xi) = 2 \operatorname{Re} \frac{N^{1/2+i\xi}}{1/2 + i\xi} = \frac{4\sqrt{N}\cos(L\xi - \arctan(2\xi))}{\sqrt{1+4\xi^2}}, \qquad \Delta_N = P_N - M_N. \tag{3.2}$$

Both finite-N multipliers are bounded and real even. Define

$$\mathcal{H}_N(u) = \frac{1}{2\pi} \int_{\mathbb{R}} (d_\Gamma - c_\Gamma - \Delta_N) |F_u|^2 \, \mathrm{d}\xi \tag{3.3}$$

on the compact core and its admitted compact H^1 products.

Proposition 3.1 (Full pole and complete remainder)**.** *For $u \in X$,*

$$\mathrm{Pole}(u) - \frac{1}{2\pi} \int_{\mathbb{R}} M_N |F_u|^2 \, \mathrm{d}\xi = T_N(u), \qquad T_N(u) = 2 \int_{\log N}^{\infty} e^{s/2} H_u(s) \, \mathrm{d}s, \tag{3.4}$$

and $|T_N(u)| \le \tau_N N_X(u)$. On the compact core,

$$q_N(u) = \mathcal{H}_N(u) + T_N(u), \qquad |\mathcal{H}_N(u) - q_\infty(u)| \le (r_N + \tau_N) N_X(u). \tag{3.5}$$

The discrepancy forms satisfy, on all of X,

$$\left| \frac{1}{2\pi} \int_{\mathbb{R}} \Delta_N |F_u|^2 \, \mathrm{d}\xi - (p_\infty(u) - \mathrm{Pole}(u)) \right| \le (r_N + \tau_N) N_X(u). \tag{3.6}$$

Proof. Absolute Fubini and $y = x + s$ give

$$\int_{\mathbb{R}} e^{s/2} H_u(s) \, \mathrm{d}s = \left(\int_{\mathbb{R}} e^{-x/2} u(x) \, \mathrm{d}x \right) \left(\int_{\mathbb{R}} e^{y/2} u(y) \, \mathrm{d}y \right) = |F_u(i/2)|^2. \tag{3.7}$$

The lower endpoint is $-\infty$, not zero. The negative half-line supplies a genuine term and cannot be discarded. On the other hand,

$$M_N(\xi) = 2 \operatorname{Re} \int_{-\infty}^{L} e^{(1/2+i\xi)s} \, \mathrm{d}s.$$

The last kernel is integrable for fixed N, and $|F_u|^2 \in L^1$. Fourier inversion for the correlation and Fubini therefore yield

$$\frac{1}{2\pi}\int_{\mathbb{R}} M_N|F_u|^2\,\mathrm{d}\xi = 2\int_{-\infty}^{L} e^{s/2}H_u(s)\,\mathrm{d}s.$$

Subtracting from twice (3.7) proves (3.4). The overlap bound and $t = e^s$ give

$$|T_N(u)| \le 80\int_L^{\infty} e^{5s/2-\pi e^s}\,\mathrm{d}s\,N_X(u) = \tau_N N_X(u).$$

Substitution into (1.7) proves the first identity in (3.5). Also

$$\frac{1}{2\pi}\int_{\mathbb{R}} \Delta_N|F_u|^2\,\mathrm{d}\xi = p_N(u) - \mathrm{Pole}(u) + T_N(u),$$

so (2.2) proves the remaining statements. □

Polarization and the Riesz representation theorem express (3.6) as bounded selfadjoint operator convergence on the real Hilbert space X, with norm error at most $r_N + \tau_N$. There is no extra polarization factor: for a bounded symmetric operator the norm is the supremum of the absolute diagonal pairing over unit vectors. This does not bound $\int|\Delta_N||F_u|^2$, and the limiting signed operator $p_\infty - \mathrm{Pole}$ is not claimed to have vanishing norm. In (3.5) only the difference of the two gamma-containing forms is X-bounded. The full pole has been consumed once by (3.4); it is not an additional resource after that identity.

For clarity, the arithmetic endpoint convention can also be read directly. Let $\Psi(t) = \sum_{n\le t}\Lambda(n)$ be right-continuous and $E(t) = \Psi(t) - t$, so $E(1) = -1$. Stieltjes integration by parts, including an atom at N, gives for $z = 1/2 + i\xi$

$$\sum_{n\le N}\Lambda(n)n^{z-1} = \frac{N^z}{z} + \frac{z-1}{z} + E(N)N^{z-1} - (z-1)\int_1^N E(t)t^{z-2}\,\mathrm{d}t. \tag{3.8}$$

Thus Δ_N is twice the real part of the last three terms. The lower endpoint $(z-1)/z$ is present. No estimate for E or prime number theorem is used in Proposition 3.1 or in the localization theorem.

4 The weighted cross-edge gain

The cubic in (1.8) obeys $b'(t) = 6t(1-t)$ on $(0,1)$. Consequently

$$0 \le \theta_R \le \pi/2, \qquad \mathrm{Lip}(\theta_R) \le 3\pi/4, \qquad \theta_R(x) \le \frac{3\pi}{2}(|x|-R)_+^2. \tag{4.1}$$

For $x, y \in \mathbb{R}$ define

$$\ell_R(x,y) = 1 - \chi_R(x)\chi_R(y) - \eta_R(x)\eta_R(y) = 1 - \cos(\theta_R(x) - \theta_R(y)), \tag{4.2}$$

$$e_R(x,y) = 1 - \chi_R(x)\chi_R(y). \tag{4.3}$$

Both lie in $[0,1]$ and vanish if both endpoints lie in $[-R,R]$. Use the following constants throughout the proof:

$$\begin{gathered} S = e^{2R}, \qquad k_R = \pi S - 2, \qquad G_R = 40Se^{-\pi(S+1)/2}, \\ B_R = \frac{288\pi^2}{e^4 k_R^4}G_R, \qquad \alpha_R = 40S^2e^{-\pi S}. \end{gathered} \tag{4.4}$$

Lemma 4.1 (Fourth-order taper bound)**.** *For all $x, y \in \mathbb{R}$ and $R \geq 1$,*

$$\ell_R(x,y)\beta(x)\beta(y) \leq B_R, \qquad e_R(x,y)\beta(x)\beta(y) \leq 2B_R. \tag{4.5}$$

For $s \geq 0$ one also has

$$\ell_R(x,x+s)\beta(x)\beta(x+s) \leq \min\left\{\frac{9\pi^2}{32}G_R s^2, B_R\right\}. \tag{4.6}$$

Either defect times $\beta(x)\beta(x+s)$ is at most $h(s)$.

Proof. If a defect is nonzero, set $\max(|x|,|y|) = R+\delta$ with $\delta > 0$. The function

$$a(t) = \sqrt{40}\exp\bigl(2t - (\pi/2)e^{2t}\bigr), \qquad t \geq 0,$$

is decreasing and has concave logarithm. Since $\beta(x) \leq a(|x|)$, the tangent bound at R gives

$$\beta(x)\beta(y) \leq a(R+\delta)a(0) \leq G_R e^{-k_R\delta}. \tag{4.7}$$

Monotonicity of the angle in $|x|$ and $1-\cos t \leq t^2/2$ give

$$\ell_R(x,y) \leq \frac{9\pi^2}{8}\delta^4, \qquad e_R(x,y) \leq \frac{\theta_R(x)^2+\theta_R(y)^2}{2} \leq \frac{9\pi^2}{4}\delta^4.$$

Now $\sup_{\delta\geq 0}\delta^4 e^{-k_R\delta} = 256/(e^4 k_R^4)$, which proves (4.5). The Lipschitz estimate in (4.1) gives $\ell_R(x,x+s) \leq 9\pi^2 s^2/32$; on a nonzero edge (4.7) also bounds the product by G_R. This proves (4.6). The final assertion follows from (2.1) and the bound of one on either defect. □

The factor k_R^{-4} records the order of the taper at the inner endpoint. The exponential factor in G_R comes from the actual large endpoint of the edge. Neither argument uses a sampled source value.

5 Exact signed localization and complete error accounting

For compact H^1 inputs define

$$I_\ell(s) = \int_{\mathbb{R}} \ell_R(x,x+s)u(x)u(x+s)\,\mathrm{d}x, \qquad I_e(s) = \int_{\mathbb{R}} e_R(x,x+s)u(x)u(x+s)\,\mathrm{d}x.$$

Expansion of the gamma squares gives exactly

$$t_\Gamma(\chi_R u) + t_\Gamma(\eta_R u) - t_\Gamma(u) = 2\int_0^\infty \rho(s)I_\ell(s)\,\mathrm{d}s. \tag{5.1}$$

Evenness in (3.7) yields

$$\mathrm{Pole}(u) = 4\int_0^\infty C(s)H_u(s)\,\mathrm{d}s. \tag{5.2}$$

The same identity holds separately for the cutoff products because the cutoffs are even. Subtracting the full forms therefore gives the direct one-cutoff identity

$$\begin{aligned} q_\infty(u) - q_\infty(\chi_R u) - t_\Gamma(\eta_R u) = {} & -c_\Gamma\|\eta_R u\|_2^2 - 2\int_0^\infty \rho(s)I_\ell(s)\,\mathrm{d}s \\ & - 2\sum_{n\geq 2} a_n I_e(\log n) + 4\int_0^\infty C(s)I_e(s)\,\mathrm{d}s. \end{aligned} \tag{5.3}$$

In particular the last term is the signed difference $\mathrm{Pole}(u) - \mathrm{Pole}(\chi_R u)$, not an extra positive allowance.

Write $u = \beta f$, so $\|f\|_2^2 = N_X(u)$. Each uniform weighted edge bound becomes a correlation bound with the same constant by $\int |f(x)f(x+s)|\,\mathrm{d}x \leq \|f\|_2^2$. We estimate the four terms in (5.3) separately, without splitting any frequency observation into positive and negative parts.

5.1 The gamma defect

Put

$$s_R^* = \frac{32}{e^2 k_R^2} < 1, \qquad L_R^* = \frac{5}{2} + \log \frac{e^2 k_R^2}{32}. \tag{5.4}$$

For $0 < s \le 1$, $\rho(s) = e^{s/2}/(2\sinh s) \le 1/s$ because $e^{1/2} < 2$ and $\sinh s \ge s$. Also

$$\int_1^\infty \rho(s)\,\mathrm{d}s \le \frac{2e^{-1/2}}{1-e^{-2}} < 2;$$

the last inequality follows from $e^{1/2} > 3/2$ and $e^2 > 4$. The two alternatives in (4.6) are equal at s_R^*. Splitting at s_R^* and at 1 gives

$$2\left|\int_0^\infty \rho(s) I_\ell(s)\,\mathrm{d}s\right| \le 2B_R\left(\frac{1}{2} + \log\frac{1}{s_R^*} + 2\right) N_X(u) = 2B_R L_R^* N_X(u). \tag{5.5}$$

5.2 Every prime power and the full pole

Let $K = \lfloor S \rfloor$. Since $\Lambda(n) \le \log n$,

$$\sum_{2\le n\le S} a_n \le (\log S) \sum_{2\le n\le S} n^{-1/2} \le 2(\log S)\sqrt{S} = 4Re^R.$$

Use $2B_R$ for the interior edge bound and $h(\log n)$ for all $n > S$. Then

$$2\left|\sum_{n\ge 2} a_n I_e(\log n)\right| \le (16Re^R B_R + r_K) N_X(u). \tag{5.6}$$

The strict tail is exactly $n > K$, including every prime power there.

For the pole defect, use $2B_R$ on $0 \le s \le 2R$ and the full overlap envelope on the remaining ray. Since $\int_0^{2R} C(s)\,\mathrm{d}s = 2\sinh R$, the first contribution is at most $8e^R B_R N_X(u)$. In the tail, $4C(s)h(s) = 80(e^{5s/2} + e^{3s/2})e^{-\pi e^s}$. With $t = e^s$, the growing part gives τ_S, while the other is at most τ_S/S. Therefore

$$4\left|\int_0^\infty C(s) I_e(s)\,\mathrm{d}s\right| \le [8e^R B_R + (1+S^{-1})\tau_S] N_X(u). \tag{5.7}$$

Finally $\eta_R = 0$ on $[-R, R]$ and the envelope decreases for $|x| \ge 0$, so $c_\Gamma \|\eta_R u\|_2^2 \le 8\alpha_R N_X(u)$.

Combining these estimates proves (1.9) with

$$\boxed{\varepsilon_R = 2B_R[L_R^* + (8R+4)e^R] + 8\alpha_R + r_{\lfloor S\rfloor} + (1+S^{-1})\tau_S.} \tag{5.8}$$

This estimates the direct identity (5.3). No standalone exterior error is added a second time. In particular no fraction of $t_\Gamma(u)$ has been spent in the error bound.

5.3 An elementary bound for all residual constants

For $n \ge 1$, $\log n \le \sqrt{n}$: the maximum of $\log t/\sqrt{t}$ on $t \ge 1$ is $2/e < 1$. Let $q = e^{-\pi} < 1/8$. The geometric second moment, with $m = K + 1$, yields

$$\sum_{n>K} n^2 q^n = m^2 q^m \left\{\frac{1}{1-q} + \frac{2q}{m(1-q)^2} + \frac{q(1+q)}{m^2(1-q)^3}\right\} < 2m^2 q^m,$$

since the bracket is at most $8/7 + 16/49 + 72/343 = 576/343 < 2$. Hence

$$r_K \le 160(K+1)^2 e^{-\pi(K+1)} \qquad (K \ge 1). \tag{5.9}$$

Because $S \ge e^2 > 4$, (2.4), (4.4) and (5.9) imply

$$8\alpha_R + r_{\lfloor S \rfloor} + (1 + S^{-1})\tau_S < (320 + 250 + 25)S^2 e^{-\pi S} < 600 S^2 e^{-\pi S}. \tag{5.10}$$

For the middle term use $K + 1 \le S + 1 < 5S/4$ and $K + 1 > S$. For the last use $\pi - 3/(2S) > 21/8$, $1 + 1/S < 5/4$ and $\sqrt{S} > 2$.

Also $k_R > (5/2)S$, $k_R < 4S$, $2 < e < 3$ and $3 < \pi < 4$ give

$$B_R < 300 S^{-3} e^{-\pi S/2}, \qquad L_R^* < \frac{9}{2} + 2\log S. \tag{5.11}$$

For the first constant, $\pi^2/e^4 < 1$ and $e^{-\pi/2} < 1$ reduce it to $11520 \cdot 16/625 = 184320/625 < 300$. Since $\sqrt{S} > 2$,

$$L_R^* + (8R + 4)e^R < 7(1 + \log S)\sqrt{S}.$$

The first term of (5.8) is therefore at most $4200(1 + \log S)S^{-5/2}e^{-\pi S/2}$. Finally $S^{9/2}e^{-\pi S/2}$ decreases for $S \ge 4$ and is less than 8 there: at 4 it is $512e^{-2\pi} < 512e^{-6} < 8$. Thus (5.10) contributes less than another $4800(1 + \log S)S^{-5/2}e^{-\pi S/2}$. This proves (1.10) and completes the proof of Theorem 1.1.

6 Support matching and the remaining signed interior

Corollary 6.1 (Matched finite-prime localization)**.** *For an integer $N \ge 81$, set*

$$R = \tfrac{1}{2}\log N - 1 \ge 1, \qquad S = N/e^2. \tag{6.1}$$

Then for $u \in \mathcal{D}_0$,

$$|\mathcal{H}_N(u) - \mathcal{H}_N(\chi_R u) - t_\Gamma(\eta_R u)| \le (\varepsilon_R + r_N + \tau_N)N_X(u), \tag{6.2}$$

$$|q_N(u) - q_N(\chi_R u) - t_\Gamma(\eta_R u)| \le (\varepsilon_R + r_N)N_X(u). \tag{6.3}$$

Proof. The localized input $v = \chi_R u$ is supported in $[-\frac{1}{2}\log N, \frac{1}{2}\log N]$. Its correlation is zero for every displacement $s \ge \log N$. At equality the possible overlap is a single point and has measure zero. Hence, exactly,

$$p_\infty(v) = p_N(v), \qquad T_N(v) = 0, \qquad q_\infty(v) = q_N(v) = \mathcal{H}_N(v). \tag{6.4}$$

This does not discard a nonzero prime-power atom at $n = N$. Use (3.5) on u only and (6.4) on v, followed by Theorem 1.1. For (6.3), use $|q_N(u) - q_\infty(u)| \le r_N N_X(u)$ instead. No second interior truncation error is needed. □

All errors in the corollary tend to zero. The leading displayed bound has the scale

$$(1 + \log N)N^{-5/2}\exp\left(-\frac{\pi N}{2e^2}\right)$$

up to fixed constants. For fixed R, any integer $K \ge \lceil e^{2(R+1)} \rceil$ gives the exact identity $q_\infty(\chi_R u) = q_K(\chi_R u)$. In both statements the full continuum of gamma shifts is retained; even shifts larger than the support diameter contribute through the difference norm in (1.4).

Remark 6.2 (Centered inputs can acquire a negative pole defect). If $\int_{\mathbb{R}} Cu = 0$, then $\mathrm{Pole}(u) = 0$, but multiplication by χ_R need not preserve that constraint. In (5.3) the pole defect is then exactly $-\mathrm{Pole}(\chi_R u) \le 0$. In particular (5.7) gives

$$\mathrm{Pole}(\chi_R u) \le [8e^R B_R + (1 + S^{-1})\tau_S] N_X(u).$$

The negative leakage has already been included in ε_R. Dropping it on the ground that the original input is centered would be incorrect.

The signed difference in (1.9) polarizes to an X-bounded symmetric operator of norm at most ε_R. This statement concerns the difference, not either unbounded gamma-containing form separately. Likewise (6.2) is not a bound $\mathcal{H}_N(u) \ge -o(1)N_X(u)$: it retains the unresolved interior term $\mathcal{H}_N(\chi_R u)$. No smallest positive interior margin is certified, and no comparison of such a margin with the uniform error is asserted.

The arguments use only elementary weighted correlation estimates, the theta relation proved below, and exact algebra. They do not use a prime number theorem, a zero-location or zero-simplicity hypothesis, a spectral projection, or a ground-state/radical identity. Their outcome is an explicit vanishing localization defect with a signed infinite-dimensional interior left intact. Neither the Riemann hypothesis nor its negation follows.

A Theta kernel and normalization

This appendix supplies the theta facts used in the localization argument, including all normalization factors. For $t > 0$ and $x \in \mathbb{R}$, define

$$\vartheta(t) = \sum_{n\in\mathbb{Z}} e^{-\pi n^2 t}, \qquad G(x) = e^{x/2}\vartheta(e^{2x}), \qquad \kappa(x) = \frac{G''(x) - G(x)/4}{2},$$
$$C(x) = \cosh(x/2), \qquad W(x) = \frac{\kappa(x)}{2C(x)}. \tag{A.1}$$

We prove that κ and W are smooth, even and strictly positive, that

$$0 < W(x) \le 40 \exp\Big(4|x| - \pi e^{2|x|}\Big), \tag{A.2}$$

and that

$$\int_{\mathbb{R}} 2C(x)\kappa(x)\,\mathrm{d}x = 1. \tag{A.3}$$

In particular, (A.3) is equivalently $\int_{\mathbb{R}} 4C(x)^2 W(x)\,\mathrm{d}x = 1$; it is not a claim that W itself has unit integral.

Gaussian periodization. For completeness, the needed theta transformation follows directly from the Gaussian. In the Fourier convention

$$F_t(\xi) = \int_{\mathbb{R}} e^{-\pi t y^2} e^{-2\pi i \xi y}\,\mathrm{d}y,$$

differentiation under the integral and integration by parts give $F_t'(\xi) = -(2\pi\xi/t)F_t(\xi)$. Also $F_t(0) = t^{-1/2}$: indeed, the square of $\int_{\mathbb{R}} e^{-\pi y^2}\,\mathrm{d}y$ is, by polar coordinates, $2\pi\int_0^\infty re^{-\pi r^2}\,\mathrm{d}r = 1$. Solving the differential equation therefore gives

$$F_t(\xi) = t^{-1/2} e^{-\pi\xi^2/t}. \tag{A.4}$$

All differentiations and integrations by parts here are justified by integrable polynomial multiples of a Gaussian.

The periodization $P_t(y) = \sum_{n\in\mathbb{Z}} e^{-\pi t(y+n)^2}$ is smooth and 1-periodic, with locally uniformly convergent differentiated series. Its kth Fourier coefficient, by an absolutely convergent change of variables, is $F_t(k)$. Thus

$$P_t(y) = t^{-1/2} \sum_{k\in\mathbb{Z}} e^{-\pi k^2/t} e^{2\pi i k y}.$$

For an explicit uniqueness justification, both sides are continuous and have the same Fourier coefficients; convolution of their difference with the Fejér kernel $K_N(y) = (N+1)^{-1} |\sum_{r=0}^{N} e^{2\pi i r y}|^2$ is zero. These convolutions converge uniformly to the difference: $K_N \geq 0$, $\int_0^1 K_N = 1$, and on every closed set away from the integers, $K_N = O((N+1)^{-1})$. Uniform continuity supplies the remaining neighborhood of the integers. Evaluating at $y = 0$ proves

$$\vartheta(t) = t^{-1/2} \vartheta(1/t). \tag{A.5}$$

Termwise differentiation of ϑ is valid on every compact subinterval of $(0, \infty)$. Differentiating (A.5) at $t = 1$ yields

$$\vartheta'(1) = -\frac{1}{4} \vartheta(1). \tag{A.6}$$

Smoothness, parity and positivity. Equation (A.5) gives $G(-x) = G(x)$, so G, κ, and W are smooth and even. Put $t = e^{2x}$. Direct differentiation of (A.1) gives

$$\kappa(x) = e^{x/2} \big(2t^2 \vartheta''(t) + 3t \vartheta'(t)\big). \tag{A.7}$$

The constant term of ϑ is annihilated. Pairing its positive and negative indices gives, in particular for $x \geq 0$,

$$\kappa(x) = \sum_{j\geq 1} \left(4\pi^2 j^4 e^{9x/2} - 6\pi j^2 e^{5x/2}\right) e^{-\pi j^2 e^{2x}}. \tag{A.8}$$

For $x \geq 0$, each coefficient is $e^{x/2}(4u^2 - 6u) > 0$, where $u = \pi j^2 e^{2x} \geq \pi > 3$. Consequently $\kappa > 0$ on that half-line, and evenness gives positivity everywhere. Since $C > 0$, the same holds for W.

A complete uniform envelope. The elementary bounds $3 < \pi < 22/7$ can be obtained without numerical input from

$$\pi = 4 \int_0^1 \frac{\mathrm{d}u}{1+u^2} > 4 \sum_{k=0}^{7} \frac{(-1)^k}{2k+1} = \frac{135904}{45045} > 3, \qquad \frac{22}{7} - \pi = \int_0^1 \frac{u^4 (1-u)^4}{1+u^2} \, \mathrm{d}u > 0.$$

The first inequality follows by the finite geometric expansion with positive remainder, and the second identity by polynomial division. Also $e^3 > \sum_{k=0}^{4} 3^k/k! = 131/8 > 16$, so $e^{-3\pi} < e^{-9} < 1/4096$.

For $x \geq 0$, use $2C(x) \geq e^{x/2}$ and discard the negative term in (A.8). Since $j^2 - 1 \geq 3(j-1)$, this gives

$$\begin{aligned} W(x) &\leq 4\pi^2 e^{4x - \pi e^{2x}} \sum_{j\geq 1} j^4 e^{-\pi(j^2-1)e^{2x}} \\ &\leq 4 \left(\frac{22}{7}\right)^2 e^{4x - \pi e^{2x}} \sum_{j\geq 1} \frac{j^4}{4096^{j-1}}. \end{aligned}$$

This bounds the full theta tail, not merely its first summand. For every integer $j \geq 1$, $j^4 \leq 16^{j-1}$, by induction using $((j+1)/j)^4 \leq 16$. Hence the coefficient above is bounded by

$$4\left(\frac{22}{7}\right)^2 \sum_{j\geq 1} \frac{j^4}{4096^{j-1}} \leq 4\left(\frac{22}{7}\right)^2 \frac{256}{255} = \frac{495616}{12495} < 40, \qquad 40 - \frac{495616}{12495} = \frac{4184}{12495} > 0.$$

Evenness proves (A.2) on the whole line. In particular, $2C\kappa = 4C^2W$ is integrable.

Exact mass. For $t \geq 1$, the differentiated theta series gives $\vartheta(t) \to 1$ and $t^a\vartheta'(t) \to 0$ for every fixed real a. For example, after factoring out $e^{-\pi t}$, the remaining sums are bounded by the convergent series $\sum_{j\geq 1} j^m e^{-\pi(j^2-1)}$, with a suitable fixed m. Using evenness, $\mathrm{d}x = \mathrm{d}t/(2t)$, and (A.7), one obtains

$$\begin{aligned}\int_{\mathbb{R}} 2C\kappa \,\mathrm{d}x &= \int_1^\infty \left[2(t^{3/2}+t)\vartheta''(t) + 3(\sqrt{t}+1)\vartheta'(t)\right] \mathrm{d}t \\ &= \Big[2(t^{3/2}+t)\vartheta'(t) + \vartheta(t)\Big]_1^\infty \\ &= 1 - 4\vartheta'(1) - \vartheta(1) = 1,\end{aligned}$$

where the last equality is exactly (A.6). The displayed theta tail bounds justify the endpoint limits and the integrations. This proves (A.3) directly, with no Mellin transform or spectral assumption.

Data availability and manuscript status

The companion manuscript source, theta appendix and source-free scalar checks are provided at `https://github.com/serabing-hash/theta-weighted-certificate/tree/main/research/2026-10-27-signed-cubic-localization`. The scalar checks verify recorded finite arithmetic; they do not certify the analytic proof. Artificial intelligence assistance was used for drafting, mathematical exposition checks and document preparation. This manuscript is a preprint and makes no claim of human peer review.